A Multimodal Prototype Correction and Multidimensional Knowledge Distillation Framework for Cross-Domain Few-Shot Hyperspectral Image Classification

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***Abstract*—Prototype-driven few-shot learning (FSL) holds promise for cross-domain hyperspectral image (HSI) classification but encounters two key limitations: 1) over-reliance on visual features overlooks deep semantic information, hindering the differentiation of visually similar categories and 2) inade quate semantic connections between local and holistic features restrict the utilization of spatial–spectral characteristics. To tackle these issues, we propose a multimodal prototype correction and multidimensional knowledge distillation framework for cross-domain few-shot hyperspectral image classification MCMD-CFSC. This framework implements a multimodal prototype correction strategy that leverages pre-trained language models to extract spatial and spectral text features from spatial and spectral text descriptions, thereby aligning image spatial prototypes with spatial text features and spectral prototypes with spectral text features to enhance the recognition of diverse ground objects. In addition, a multidimensional knowledge distillation mechanism is developed to ensure semantic consistency between local and holistic features across spatial–spectral dimensions, bolstering the model's discriminative power and generalization ability. Experimental results demonstrate that MCMD-CFSC significantly surpasses existing deep learning and FSL techniques on four hyperspectral datasets. The code is available at https://github.com/Li-ZK/MCMD-CFSC-2025.**

***Index Terms*—Few-shot learning (FSL), hyperspectral image (HSI) classification, multidimensional knowledge distillation (MD), multimodal prototype correction (MC).**

I. Introduction

HYPERSPECTRAL images (HSIs) constitute a 3-D data cube, integrating spectral and spatial information to characterize the radiation properties and geometric relationships of ground objects [1], [2]. Owing to their capacity for detailed spectral differentiation and rich information content, HSIs have found extensive applications in diverse fields, including mineral identification, military target detection, and precision agriculture [3], [4], [5], [6].

The classification of HSIs is a critical task in HSI analysis. Early research predominantly relied on traditional machine learning algorithms, such as k-nearest neighbor (k-NN) [7] and support vector machines (SVMs) [8], which primarily focused on leveraging spectral information for classification tasks. Subsequent advancements introduced more sophisticated methods, including composite kernels [9] and joint sparse representation [10], [11], which aimed to enhance classification performance by incorporating additional feature representations. However, these approaches often necessitated manually designed features, limiting their adaptability across varying application scenarios.

In recent years, deep learning has revolutionized HSI classification, achieving remarkable performance improvements [12], [13], [14], [15]. Architectures such as 3-D convolutional neural networks (3DCNNs) [16] and spectral–spatial residual network (SSRN) [17] have emerged as pivotal frameworks in this domain. Specifically, 3DCNN excels by concurrently capturing spatial and spectral features through 3-D convolution operations, making them highly suitable for hyperspectral data with inherent spectral dimensionality. Meanwhile, SSRN introduces residual structures to mitigate the vanishing gradient problem in deep networks, facilitating more profound feature learning and significantly enhancing classification accuracy. These advancements underscore the transformative potential of deep learning in HSI classification. Despite the notable achievements attained using large labeled datasets, the high costs associated with HSI annotation pose significant challenges for practical applications. Consequently, few-shot HSI classification has emerged as a prominent area of research, which aims to learn from source (base) classes that have sufficient labeled data and then adapt to target (novel) classes with limited labeled samples has emerged as a prominent area of research [18], [19], [20], [21]. To tackle this issue, researchers have increasingly turned to metalearning approaches [22], [23], [24], [25], [26], [27], [28], [29], which focus on learning a model from a diverse set of known tasks that

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can rapidly adapt to new tasks using minimal labeled data. In the training phase, the dataset is partitioned into various metatasks to extract metaknowledge, thereby enhancing the model's ability to generalize to novel classes. During the testing phase, classification can be performed without altering the previously trained model. Metalearning methodologies can be primarily classified into three categories: model-based, optimization-based, and metric-based methods. Among these, metric-based approaches have gained popularity in few-shot hyperspectral classification due to their straightforwardness and efficacy [30], [31]. For instance, Li et al. [32] proposed a deep cross-domain FSL (DCFSL) framework that augments model generalization in unseen conditions by incorporating a domain adaptation module within the metric learning process and employing adversarial training to mitigate disparities in feature distributions between source and target domains. Similarly, Guo et al. [33] introduced a distribution-aware metric learning technique that comprehensively models class distribution characteristics by integrating both sample mean information and intraclass covariance matrices into the metric calculations. Li et al. [34] advanced an adaptive prototype update (APU) mechanism, which enhances model stability by dynamically modifying class prototype representations to accommodate fluctuations in sample distributions. Furthermore, Zhang et al. [35] proposed an innovative graph information aggregation cross-domain FSL (Gia-CFSL) framework that utilizes both feature-level and distribution-level cross-domain graph alignment strategies to alleviate issues related to domain shift.

Current metric-based approaches primarily concentrate on visual features while often overlooking the potential of multimodal complementary information [36]. This oversight significantly hinders the ability to discriminate between spectrally similar land cover categories. Humans inherently integrate multimodal cues—such as visual and linguistic information—when recognizing unfamiliar objects, a cognitive capability that allows for accurate category identification even with minimal examples. This mechanism of multimodal information fusion offers valuable insights for tackling challenges in few-shot classification. Recent studies have introduced text modality into few-shot HSI classification. By integrating the semantic information from text with the visual information from images, this approach can effectively compensate for the limitations of single modalities in feature representation. For instance, Huang and Choi [37] introduced a dual-stream network architecture that facilitates the dynamic fusion of visual and textual features through attention mechanisms. Xing et al. [38] proposed an adaptive modality mixing mechanism (AM3), which leverages category word vectors to guide the learning of category prototypes. In addition, Fortin and Chaib-draa [39] developed contextual representations for each category based on attention mechanisms, thereby enhancing the model's expressive capabilities through the adaptive fusion of textual and visual prototypes.

In the context of cross-domain few-shot hyperspectral classification, Li et al. [40] explored the use of textual semantics to direct the model in acquiring generalizable visual features. However, this approach relies solely on basic category labels as textual descriptors, which limits the representation of the intricate spatial structure and fine-grained spectral response characteristics inherent to hyperspectral data. Consequently, such coarse-grained semantic information can hinder the model's ability to achieve a deeper understanding and more precise recognition of land cover categories.

Hyperspectral data contains extensive multidimensional information, and enhancing classification performance relies on effectively extracting and utilizing this data. In spatial–spectral feature fusion, several innovative methods have been proposed: Sun et al. [41] introduced a transformer-based feature fusion framework (SSFT) that employs convolutional networks to extract shallow spatial–spectral features, followed by deep feature learning and fusion using Transformers; Wang et al. [42] developed an adaptive spatial–spectral mixing network (ASSMN) that adaptively fuses spatial–spectral features by assessing their importance; and Asker and Güngör [43] presented a deeply separable squeeze-and-excitation network (DSC-DSENet) that efficiently combines 3D-DSC and DSENet for joint extraction of spatial–spectral features. However, these methods fail to account for the deep semantic relationships between local and holistic features, limiting the full exploitation of hyperspectral data's rich multidimensional characteristics.

To address the aforementioned challenges, the article proposes a multimodal prototype correction and multidimensional knowledge distillation framework for cross-domain few-shot hyperspectral image classification MCMD-CFSC. This approach effectively refines the spatial–spectral prototypes of images by aligning them with meticulously extracted spatial and spectral textual features. It harnesses the robust prior knowledge from large language models to thoroughly extract structural information related to land cover from spatial text and spectral characteristics from spectral text. In addition, the method establishes semantic consistency between local and holistic features of samples across spatial–spectral dimensions, enabling the extraction of more discriminative spatial–spectral features. Moreover, the method implements an exponential moving average (EMA) strategy to dynamically update the network parameters, thereby accumulating cross-episode knowledge to enhance the model's capacity for knowledge transfer and improve its generalization performance.

The main contributions of this work are summarized as follows.

1) A novel cross-domain few-shot HSI classification framework is proposed, which integrates multimodal prototype correction and multidimensional knowledge distillation. By leveraging innovative multimodal fusion and feature learning strategies, significant improvements in classification performance are achieved under sample-scarce conditions.
2) A multimodal prototype correction strategy is developed that utilizes spatial and spectral text features to precisely correct image spatial and spectral prototypes, effectively mitigating interclass similarity interference and augmenting the model's capability for distinguishing diverse land cover types.

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3) An innovative multidimensional knowledge distillation mechanism is conceived, designed to establish semantic consistency between local and holistic sample features across spatial–spectral dimensions. Through this approach, the model's discriminative capabilities and generalization performance are substantively enhanced, addressing key challenges in HSI classification.

The article is organized as follows. Section II introduces foundational concepts of cross-domain few-shot learning (FSL), text-assisted image recognition, and knowledge distillation. Section III elaborates on the proposed MCMD-CFSC method. Section IV presents comprehensive experimental results and analyses. Section V draws conclusions.

II. Related Work

A. Cross-Domain FSL

Hyperspectral remote sensing technology has made significant strides, finding wide application in various fields such as agriculture, environmental monitoring, and mineral exploration. Consequently, cross-domain few-shot HSI classification has emerged as a research frontier. The core motivation for this research direction stems from a severe real-world challenge: comprehensive data collection and detailed annotation in a target scene are often prohibitively expensive and time-consuming, making it difficult to obtain sufficient labeled samples. Therefore, how to effectively transfer knowledge learned from a data-rich source domain to assist in the classification task in a target domain with only a few labeled samples has become a key research topic. Formally, the cross-domain FSL scenario is defined as follows: given a source domain $D_s = X_s, Y_s$ with abundant labeled data and a target domain $D_t = X_t, Y_t$. In the target domain, each class has only a very small number of labeled samples (few-shot), and the class sets of the source and target domains are disjoint (cross-domain). The primary challenge in this scenario lies in simultaneously coping with the dual predicament of significant domain distribution differences caused by factors like different sensors, illumination, or geographical locations, and the extreme scarcity of available samples in the target domain. In recent years, various methods have emerged to address these challenges. For instance, Hu et al. [44] proposed a dual-tuning metalearning algorithm, while Wang et al. [45] developed a class-attention-based method; both aim to learn more discriminative and domain-invariant features. Similarly, the adaptive subspace method introduced by Bai et al. [46] also strives to mitigate the cross-domain problem. These methods have achieved notable success in directly aligning visual feature distributions or enhancing domain invariance. However, they primarily rely on visual modality information, to some extent overlooking the immense potential of introducing external knowledge (such as text semantics) to enrich feature representations. The offers new possibilities for enhancing a model's discriminative ability and generalization performance in complex scenarios. The article constructs more robust feature representations through multimodal prototype correction and multidimensional knowledge distillation to better generalize to the sample-scarce target domain.

B. Text-Assisted Image Recognition

In the field of image classification, fusing textual and visual information has become a key research direction for enhancing model performance. A large body of research has shown that introducing semantic knowledge from the text modality can provide rich contextual information to the model, thereby significantly strengthening its classification ability [47], [48], [49], especially when distinguishing between categories that are visually similar but semantically different [50]. Inspired by this, recent studies have begun to introduce textual information into few-shot HSI classification. For example, Huang and Choi [37] designed a dual-stream network to dynamically fuse visual and textual features, while Xing et al. [38] utilized class word vectors to guide prototype learning. However, current text-assisted HSI classification methods suffer from a core limitation: they mostly rely on coarse-grained textual information (such as using only class names) and simple fusion strategies (such as feature concatenation). This approach neglects the inherent complexity of hyperspectral data—that is, land cover features not only have class distinctions but also possess unique spatial structures (e.g., texture and shape) and fine-grained spectral response characteristics. Using only a single piece of semantic information, like a class label, is insufficient to fully guide the model in learning features that can distinguish subtle differences. Therefore, a key research challenge is: how to design innovative methods that leverage fine-grained text semantics, matching the spatial and spectral properties of hyperspectral data to guide efficient feature learning under the condition of scarce labeled samples, thereby achieving more accurate classification. This article proposes a multimodal prototype rectification strategy, designing fine-grained textual descriptions for both spatial and spectral dimensions. It utilizes these text features to precisely correct the spatial and spectral prototypes of the image, achieving a deeper level of modal fusion.

C. Knowledge Distillation

Knowledge distillation, as a powerful technique for model compression and knowledge transfer, has garnered significant attention in the field of deep learning [51], [52]. Its core idea is to have a smaller "student" model learn by imitating the output (e.g., soft labels) or intermediate features of a larger "teacher" model, thereby achieving performance comparable to the teacher model with significantly fewer parameters [53]. With deepening research, various innovative knowledge distillation methods have emerged successively. Relational knowledge distillation transfers structured knowledge by having student models learn intersample distance matrices from teacher models [54]. Attention transfer methods guide students to focus on key spatial regions by conveying teacher attention patterns [55]. Cross-view knowledge distillation leverages multiperspective or multimodal information to provide richer knowledge transfer, improving model generalization in complex scenarios [56]. Inspired by these knowledge distillation techniques, researchers in the HSI classification field have also actively explored various knowledge distillation methods. Yue et al. [57] proposed a self-supervised

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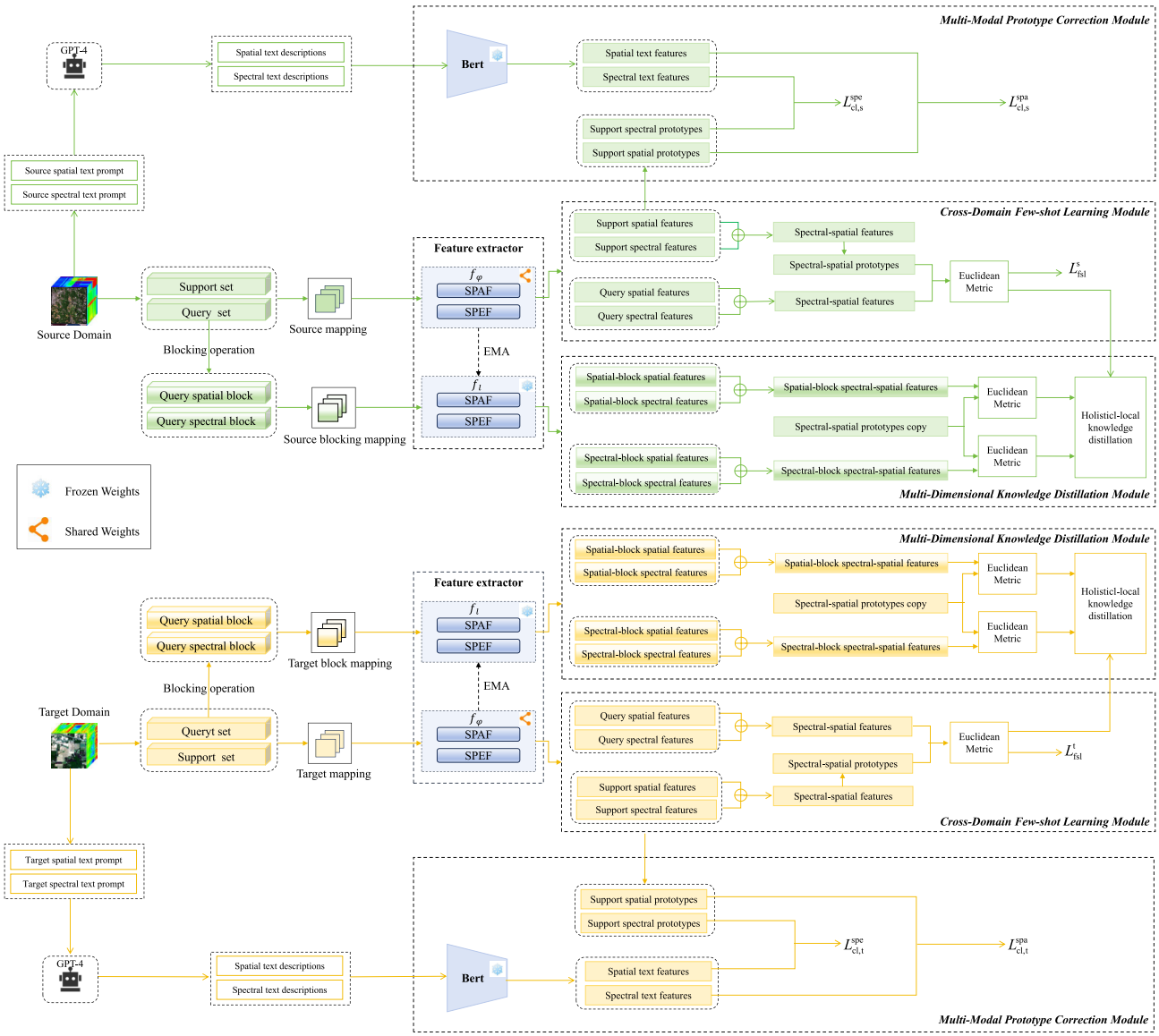


Fig. 1. Framework of the proposed MCMD-CFSC method, consisting of five modules: image feature extraction, text feature extraction, source and target domain FSL, MC, and MD modules.

learning framework based on adaptive distillation, which calculates sample similarity through spatial–spectral joint distance and generates adaptive soft labels for knowledge distillation training. Chi et al. [58] designed a self-supervised knowledge distillation method (SSKD) that performs similarity discrimination by integrating spatial and spectral distances, adopting an adaptive nearest neighbor matching strategy to generate soft labels for unlabeled samples, thereby mining more classification-valuable information. Feng et al. [59] developed a cross-domain FSL method based on decoupled knowledge distillation (DKD-FSL), which decomposes knowledge distillation into task-relevant knowledge and data-intrinsic knowledge through a decoupling interaction module, and designs a discriminative information refinement module to extract key discriminative features from redundant hyperspectral information, effectively addressing significant distribution difference issues in cross-domain scenarios. However, they often fail to fully exploit the deep semantic correlation between a sample's local features and its global features. This correlation is crucial for forming a comprehensive and highly discriminative feature representation. Therefore, a key challenge is how to design a novel knowledge distillation mechanism that can, within the model and across spatial–spectral dimensions, enforce semantic consistency between local and global features, thereby prompting the model to learn more discriminative and generalizable feature representations. The article proposes a multidimensional knowledge distillation mechanism that forces the model to learn semantic consistency between local and global features by distilling the knowledge of a sample's global feature (teacher) to its corresponding local features (students). This mechanism enables the model to extract highly discriminative

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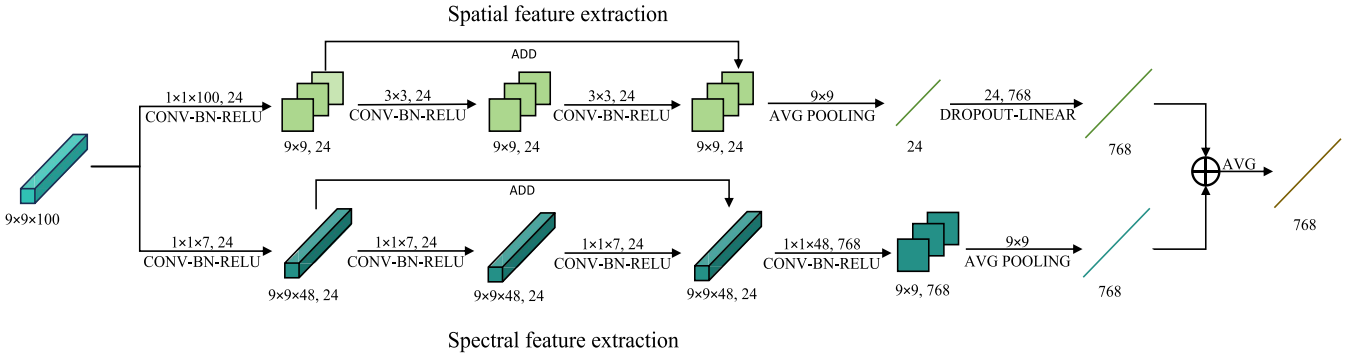


Fig. 2. Architecture of image feature extractor.

spatial–spectral features from hyperspectral data that encompass both detail and a holistic perspective.

III. Proposed Method

The proposed MCMD-CFSC framework, depicted in Fig. 1, integrates five key modules: image and text feature extraction, FSL, multimodal prototype correction (MC), and multidimensional knowledge distillation (MD). The feature extraction module uses mapping layers to align source and target domain spectral dimensions and a dual-channel embedding extractor with spatial and spectral feature channels. Utilizing Base bidirectional encoder representations from transformer (BERT), the text feature extraction module complements the image analysis. The FSL module defines classification subtasks across domains to discover shared knowledge, while the multimodal prototype correction module refines image prototypes using text description features. Finally, the multidimensional knowledge distillation module establishes semantic consistency between local and holistic sample features, enhancing the model's ability to learn discriminative and generalizable representations.

A. Image Feature Extractor

The feature extraction module comprises three key components: a mapping layer, a spatial feature extraction channel, and a spectral feature extraction channel. The architectures of the spatial and spectral feature extraction channels are shown in Fig. 2.

1) Mapping Layer: To address the differing spectral channel dimensions between the source and target domain data, a mapping layer is implemented to standardize these dimensions. This layer consists of a 2-D convolutional structure with d convolution kernels, each sized $1 \times 1 \times c$, where c and d represent the channel dimensions of the input and output data, respectively. After passing through the mapping layer, the feature maps from both domains achieve a uniform channel dimension of d.

2) Spatial Feature Extraction Channel: The spatial feature extraction channel is designed to capture the spatial neighborhood information around central pixels. It includes convolutional layers, residual blocks, and fully connected layers. The convolutional layer utilizes a $1 \times 1 \times 100$ kernel size to compress multichannel spectra into a single channel, enabling the network to focus more on spatial details. The residual blocks consist of two convolutional layers with $1 \times 3 \times 3$ kernels, effectively extracting spatial information from the central pixels. After passing through the residual blocks, the spatial data is aggregated using 9×9 global average pooling to create feature vector representations. A fully connected layer then projects these feature vectors into 768 dimensions, aligning them with the spectral dimension.

3) Spectral Feature Extraction Channel: The spectral feature extraction channel focuses on obtaining the spectral information of central pixels. It includes two convolutional layers and a residual block. The first convolutional layer, with a $1 \times 1 \times 9$ kernel size, removes redundant spectral features. The residual block features two convolutional layers, also with $1 \times 1 \times 9$ kernels, for accurate spectral information extraction. The second convolutional layer, using a $1 \times 1 \times 48$ kernel, transforms features and reduces dimensionality. Ultimately, a 768-D spectral feature vector is produced via 9×9 global average pooling.

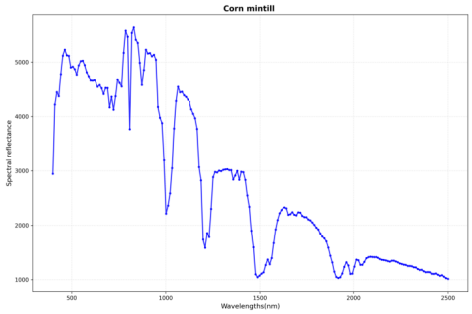
4) Spatial–Spectral Feature Fusion Module: The spatial–spectral feature fusion module employs an equal-weight fusion approach to combine spatial and spectral feature vectors, resulting in a final spatial–spectral feature vector that enhances the integration of information, increases category distinctions, and improves the network's classification performance.

B. Text Feature Extractor

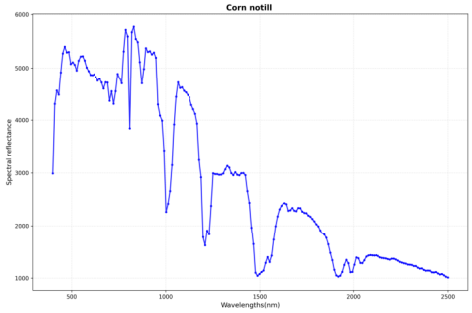
The Base BERT model [60] is chosen as the text feature extractor for the feature extraction module. BERTs are a pre-trained model built on the Transformer architecture that excels in natural language processing tasks. It learns rich semantic representations and language structures through pre-training on extensive corpora, providing a strong foundation for extracting prior knowledge in spatial–spectral description texts.

1) Input Text Preprocessing: Clean the input spatial–spectral description texts by removing special characters, punctuation, and unnecessary whitespace to enhance text quality and processing efficiency. Utilize a word segmentation tool compatible with the Base BERT model to segment the cleaned texts according to semantic and grammatical rules, converting them into word sequences for

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(a)



(b)

Fig. 3. Spectral curves of corn-mintill and corn-notill. (a) Spectral curves of corn-mintill. (b) Spectral curves of corn-notill.

better comprehension by the model. Finally, leverage the Base BERT model's built-in word vector generation mechanism to create word vector representations for each segmented word, capturing their semantic features and contextual information in a low-dimensional vector space based on pre-trained distributed semantic representations.

2) Text Feature Extraction: The Transformer encoder in the Base BERT model performs multilayer encoding on input word vector sequences. Each layer employs a multihead attention mechanism to capture relationships between words, compute attention distributions across different semantic subspaces, and update word representation vectors accordingly. This mechanism allows simultaneous focus on various semantic aspects, such as grammatical structures and associations, thereby producing richer text feature representations. After multiple encoding layers, the Base BERT model outputs the hidden layer representation vectors for each word. We integrate these vectors in specific ways, such as using the vector corresponding to the [CLS] (classification token) as the overall text feature representation. This [CLS] vector, optimized for text classification tasks during pre-training, encapsulates the overall semantic information of the text, serving as a feature representation of the spatial and spectral description text's prior knowledge, and the Base BERT model is completely frozen throughout the entire training process.

3) Text Description Generation: We generated spatial and spectral text descriptions through GPT-4. By first providing the dataset name and the name of each category, we asked GPT-4 to generate spatial texture descriptions and spectral reflectance descriptions for each category. For example, the grassland category is spatially described as "exhibits uniform and dense texture with subtle variations in height and density,"

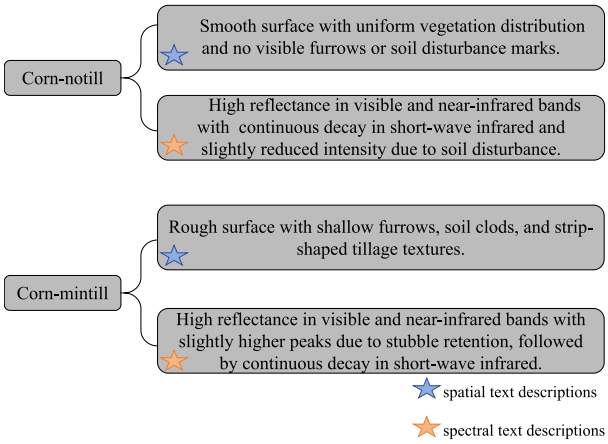


Fig. 4. Text descriptions of corn-notill and corn-mintill.

while spectrally it is characterized as "shows strong reflectance in the green band of visible light, high reflectance in the infrared band, and significantly reduced reflectance in the red-edge region."

C. Cross-Domain FSL Module

To effectively leverage knowledge from both source and target domains, we employ an episodic training strategy based on metalearning principles. This approach enables the shared feature extractor to learn discriminative representations that generalize well across domains.

Our metalearning framework operates simultaneously in both source and target domains. Each episode constitutes a metatask that simulates an FSL scenario, allowing the model to learn adaptation strategies that can be applied to novel classes.

The metalearning process begins by computing class prototypes from the support set. For each class k, its prototype $\boldsymbol{c}_k$ is calculated as the mean feature representation of its support samples

$$\boldsymbol{c}_k = \frac{1}{K} \sum_{x \in S_k^s} f_\theta(x) \tag{1}$$

where f_θ represents our feature extraction network with parameters θ, and S_k^s denotes the support samples belonging to class k.

For the classification of query samples, we employ a metric-based approach. Given a query sample x, we compute its probability of belonging to class k as follows:

$$P(y = k \,|\, x) = \frac{\exp\left(-d\left(f_\theta(x), \boldsymbol{c}_k\right)\right)}{\sum_{k=1}^{N} \exp\left(-d\left(f_\theta(x), \boldsymbol{c}_k\right)\right)} \tag{2}$$

where $d(\cdot, \cdot)$ denotes the Euclidean distance metric. This formulation encourages the feature space to be discriminative, with samples clustering around their respective class prototypes.

The FSL objective for a single episode in the source domain is defined as the negative log-likelihood

$$L_{\text{fsl}}^s = E_{S_s, Q_s}\left[-\sum_{(x,y) \in Q_s} \log p(y = k \mid x) \right]. \tag{3}$$

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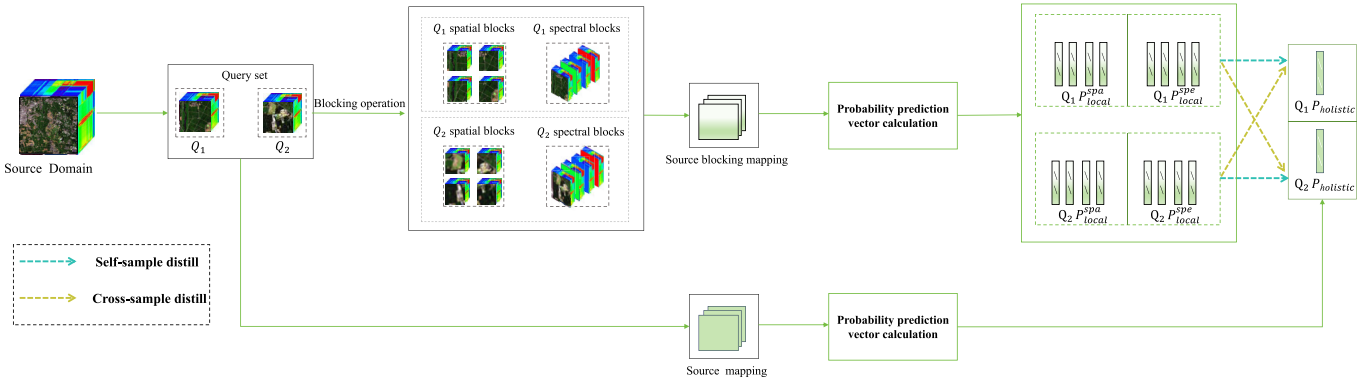


Fig. 5. Schematic of multidimensional knowledge distillation module, $Q1$ and $Q2$ belong to the same category.

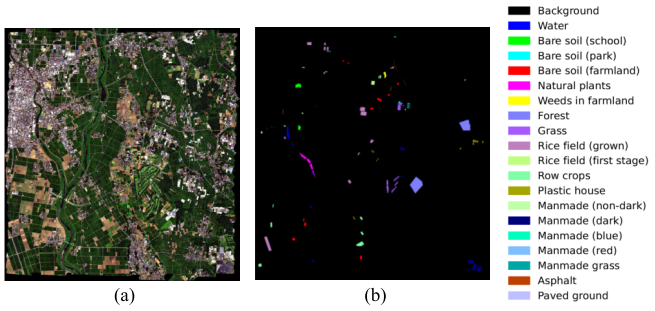


Fig. 6. Pseudo-color image and ground-truth map of CH. (a) Pseudo-color image. (b) Ground-truth map.

We apply the same episodic training procedure in the target domain, resulting in a parallel loss L^t_{fsl}. The combined FSL objective is then

$$L_{\text{fsl}} = L^s_{\text{fsl}} + L^t_{\text{fsl}}. \tag{4}$$

D. Multimodal Prototype Correction Module

In HSI classification, relying exclusively on visual features presents considerable challenges when discriminating between spectrally similar categories. Consider the recognition of "corn-notill" and "corn-mintill" as an example, as demonstrated in Fig. 3, both categories display nearly iden tical reflectance characteristics across multiple spectral bands, making differentiation extremely challenging when relying solely on visual features. This limitation becomes particularly pronounced in sample-scarce scenarios, where insufficient training data further constrains the discriminative capacity of visual-only approaches.

The textual modality, serving as a rich semantic carrier, provides crucial supplementation to overcome these visual limitations. Through carefully constructed textual descriptions in Fig. 4, the critical differences between "Corn-no-till" and "Corn-min-till" can be precisely characterized, demonstrating the essential role of integrating text semantic knowledge to achieve robust classification performance.

To leverage the complementary advantages of visual and textual modalities, we propose an innovative prototype correction strategy. This approach utilizes spatial text features extracted from spatial descriptions to align image spatial prototypes, while employing spectral text features derived from spectral descriptions to align image spectral prototypes.

Formally, in the source domain for each episode, we define T^{spa}_s and T^{spe}_s as the sets of spatial and spectral text descriptions. For each class $i \in N$, we denote the spatial and spectral text descriptions as $t^{\text{spa}}_{i,s}$ and $t^{\text{spe}}_{i,s}$, respectively. We utilize the Base BERT model as our text feature extractor f_{θ_t}, initialized with BERT-uncased weights which remain frozen during training. The text descriptions are processed through f_{θ_t} to generate text features $\boldsymbol{s}^{\text{spa}}_{i,s}$ and $\boldsymbol{s}^{\text{spe}}_{i,s}$, which encapsulate the semantic information of spatial and spectral characteristics for each class. These text features are then aligned with their corresponding image-based spatial and spectral prototypes through contrastive learning.

In our framework, we establish relationships between image prototypes and their corresponding spatial and spectral text features through contrastive learning. For each category, image prototypes and text features from the same class form positive pairs, while those from different classes serve as negative pairs. Specifically, spatial image prototypes $\boldsymbol{c}^{\text{spa}}_{i,s}$ pair with spatial text features $\boldsymbol{s}^{\text{spa}}_{i,s}$, and spectral image prototypes $\boldsymbol{c}^{\text{spe}}_{i,s}$ pair with

TABLE I
NUMBER OF SAMPLES FOR EACH CLASS IN THE CH DATASET

Class	Name	Number of Samples
1	Water	2345
2	Bare soil (school)	2859
3	Bare soil (park)	236
4	Bare soil (farmland)	48525
5	Natural plants	4297
6	Weeds in farmland	1108
7	Forest	20516
8	Grass	6515
9	Rice field (groim)	13369
10	Rice field (first stage)	1268
11	Row crops	5961
12	Plastic house	2193
13	Manmade (nan-dark)	1220
14	Manmade (dark)	7664
15	Manmade (blue)	431
16	Manmade (red)	222
17	Manmade grass	1040
18	Asphalt	801
19	Paved ground	145
Total		77592

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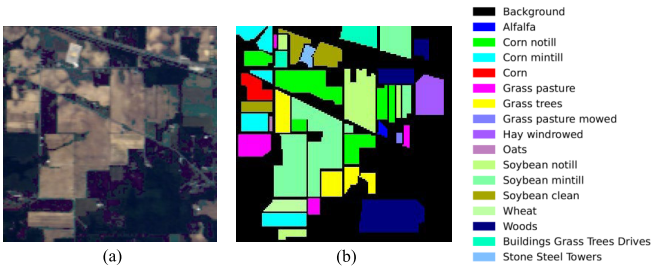


Fig. 7. Pseudo-color image and ground-truth map of IP. (a) Pseudo-color image. (b) Ground-truth map.

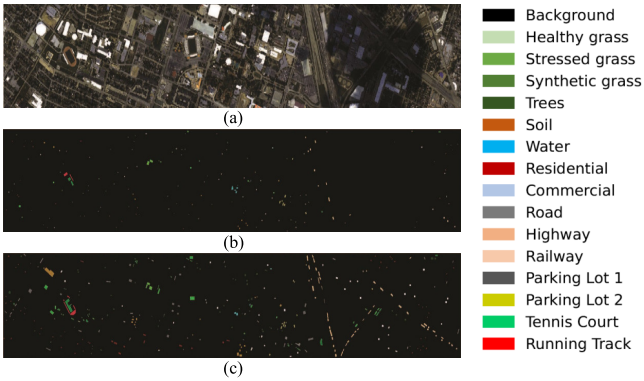


Fig. 8. Pseudo-color image and ground-truth map of HT. (a) Pseudo-color image. (b) Ground-truth map of training set. (c) Ground-truth map of testing set.

TABLE II

NUMBER OF SAMPLES FOR EACH CLASS IN THE IP DATASET

Class	Name	Number of Samples
1	Alfalfa	46
2	Corn notill	1428
3	Corn mintill	830
4	Corn	237
5	Grass pasture	483
6	Grass trees in farmland	730
7	Grass pasture mowed	28
8	Hay windrowed	478
9	Oats	20
10	Soybean notill	972
11	Soybean mintill	2455
12	Soybean clean	593
13	Wheat	205
14	Woods	1265
15	Buildings Grass Trees Drives	386
16	Stone Steel Towers	93
Total		10249

spectral text features $\boldsymbol{s}_{i,s}^{\text{spe}}$ for the ith class. Through contrastive learning, we optimize these representations by minimizing the distance between positive pairs while maximizing the distance between negative pairs. The contrastive learning objective is formulated as follows:

$$L_{\text{cl}}\left(\boldsymbol{c}_i, \boldsymbol{s}_i\right) = -\log \frac{e^{(d(\boldsymbol{c}_i, \boldsymbol{s}_i)/\tau)}}{\sum_{k=1, k\neq i}^{N} e^{(d(\boldsymbol{c}_i, \boldsymbol{c}_k)/\tau)} + \sum_{k=1}^{N} e^{(d(\boldsymbol{c}_i, \boldsymbol{s}_k)/\tau)}} \tag{5}$$

where $d(\cdot,\cdot)$ measures the Euclidean distance, and τ represents the temperature parameter controlling the influence of negative samples. The contrastive learning objectives for the spatial and



Fig. 9. Pseudo-color image and ground-truth map of SA. (a) Pseudo-color image. (b) Ground-truth map.

TABLE III

NUMBER OF TRAINING AND TESTING SET SAMPLES FOR EACH CLASS IN THE HT DATASET

Class	Name	Number of Training Set Samples	Number of Testing Set Samples
1	Healthy grass	198	1053
2	Stressed grass	190	1064
3	Synthetic grass	192	505
4	Trees	188	1056
5	Soil	186	1056
6	Water	182	143
7	Residential	196	1072
8	Commercial	191	1053
9	Road	193	1059
10	Highway	191	1036
11	Railway	181	1054
12	Parking Lot 1	192	1041
13	Parking Lot 2	184	285
14	Tennis Court	181	247
15	Running Track	187	473
Total		2832	12197

spectral components are formulated as follows:

$$L_{cl,s}^{\text{spa}} = L_{cl}\left(\boldsymbol{c}_{i,s}^{\text{spa}}, \boldsymbol{s}_{i,s}^{\text{spa}}\right) \tag{6}$$

$$L_{cl,s}^{\text{spe}} = L_{cl}\left(\boldsymbol{c}_{i,s}^{\text{spe}}, \boldsymbol{s}_{i,s}^{\text{spe}}\right). \tag{7}$$

Since the contrastive loss is asymmetric ($L_{\text{cl}}^{s}(\boldsymbol{c}_{i,s}, \boldsymbol{s}_{i,s}) \neq L_{\text{cl}}^{s}(\boldsymbol{s}_{i,s}, \boldsymbol{c}_{i,s})$), we compute the source domain episode loss as

$$L_{\text{cl}}^{s}\left(\boldsymbol{c}_{i,s}, \boldsymbol{s}_{i,s}\right) = L_{\text{cl},s}^{\text{spa}} + L_{\text{cl},s}^{\text{spe}} \tag{8}$$

$$L_{\text{mc}}^{s} = \frac{1}{2N}\sum_{i=1}^{N}\left(L_{\text{cl}}^{s}\left(\boldsymbol{c}_{i,s}, \boldsymbol{s}_{i,s}\right) + L_{\text{cl}}^{s}\left(\boldsymbol{s}_{i,s}, \boldsymbol{c}_{i,s}\right)\right). \tag{9}$$

The final contrastive loss combines both source and target domain losses

$$L_{\text{mc}} = L_{\text{mc}}^{s} + L_{\text{mc}}^{t}. \tag{10}$$

E. Multidimensional Knowledge Distillation Module

In the field of HSI classification, effectively extracting and utilizing multidimensional information from hyperspectral data is key to improving classification performance. In the spatial dimension, focusing solely on the holistic sample may overlook local texture details, while concentrating only

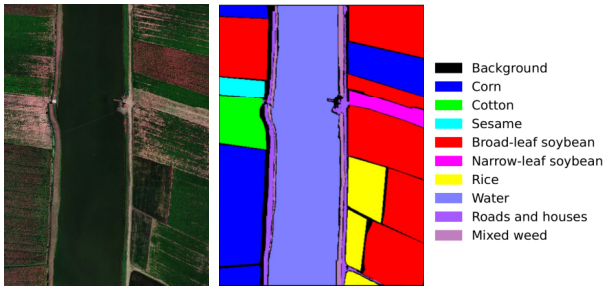
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TABLE IV
NUMBER OF SAMPLES FOR EACH CLASS IN THE SA DATASET

Class	Name	Number of Samples
1	Brocoli green weeds 1	2009
2	Brocoli green weeds 2	3726
3	Fallow	1976
4	Fallow rough plow	1394
5	Fallow smooth	2678
6	Stubble	3959
7	Celery	3579
8	Grapes untrained	11271
9	Soil vinyard develop	6203
10	Corn senesced green weeds	3278
11	Lettuce romaine 4wk	1068
12	Lettuce romaine 5wk	1927
13	Lettuce romaine 6wk	916
14	Lettuce romaine 7wk	1070
15	Vinyard untrained	7268
16	Vinyard vertical trellis	1807
Total		54129

TABLE V
NUMBER OF SAMPLES FOR EACH CLASS IN THE LK DATASET

Class	Name	Number of Samples
1	Corn	34511
2	Cotton	8374
3	Sesame	3031
4	Broad-leaf soybean	63212
5	Narrow-leaf soybean	4151
6	Rice	11854
7	Water	67056
8	Roads and houses	7124
9	Mixed weed	5229
Total		204542



(a) (b)

Fig. 10. Pseudo-color image and ground-truth map of LK. (a) Pseudo-color image. (b) Ground-truth map.

on local regions may lose the holistic structural information. Similarly, in the spectral dimension, analyzing only the complete spectral sequences may fail to capture subtle band variations, while examining only local bands might miss the holistic spectral response characteristics. For example, in differentiating between "soybean notill" and "soybean mintill," both crops exhibit similar regular planting patterns in the spatial dimension and share comparable complete spectral sequences. In such scenarios, accurate classification demands the integration of local spatial texture features with distinctive spectral responses within specific band ranges. Furthermore, when identifying crops at various growth stages, it becomes essential to jointly leverage both the overall spatial distribution patterns and the temporal variations in growth states

TABLE VI
PARAMETER TUNING OF THE MOMENTUM m USING THE TARGET DOMAIN DATASETS

Dataset	Momentum m				
	0.9	0.93	0.95	0.97	0.99
IP	82.79	82.99	**83.25**	82.75	83.13
HT	**77.51**	77.37	76.9	77.28	77.09
SA	95.15	**95.32**	95.21	95.24	95.21
LK	**97.19**	97.18	97.16	97.18	97.18

TABLE VII
PARAMETER TUNING OF THE L_{mc} WEIGHT λ_1 USING THE TARGET DOMAIN DATASETS

Dataset	L_{mc} Weight λ_1				
	0.1	0.5	1	1.5	2
IP	81.55	82.51	83.07	83.19	**83.25**
HT	**77.51**	76.26	75.99	76.07	76.02
SA	94.7	94.9	94.98	95.16	**95.32**
LK	**97.19**	97.09	96.89	96.81	96.68

TABLE VIII
PARAMETER TUNING OF THE L_{md} WEIGHT λ_2 USING THE TARGET DOMAIN DATASETS

Dataset	L_{md} Weight λ_2				
	0.1	0.5	1	1.5	2
IP	82.79	82.86	**83.25**	83	83.07
HT	76.65	76.87	**77.51**	77.23	77.12
SA	94.95	95.07	**95.32**	95.09	95.07
LK	97.15	97.16	**97.19**	97.14	97.17

manifested in complete spectral sequences. To effectively harness this multidimensional information, we propose a multidimensional knowledge distillation module. This module comprises two complementary branches: a holistic branch and a local branch. The holistic branch specializes in extracting overall structural features from complete image patches, whereas the local branch performs patch blocking to meticulously mine local detail information. The knowledge distillation process is shown in Fig. 5.

Specifically, for an image x_{q_i} in the query set, the holistic branch processes it through the feature extractor f_φ to obtain holistic features and computes the holistic prediction probability vector $\boldsymbol{p}_{q_i}$ according to (2). The local branch performs blocking on x_{q_i} along both spatial and spectral dimensions to generate block images $x^{\text{spa}}_{q_{i,r}}$ and $x^{\text{spe}}_{q_{i,r}}$, where $r \in [1, R]$ denotes the number of blocks per sample. These blocks are subsequently fed into the feature extractor f_l to obtain block feature vectors, termed as local features. Notably, f_l and f_φ share identical network architectures and initialization weights. The local prediction probability vectors $\boldsymbol{p}^{\text{spa}}_{q_{i,r}}$ and $\boldsymbol{p}^{\text{spe}}_{q_{i,r}}$ are then computed according to (2). To facilitate comprehensive utilization of information from different dimensions, we impose consistency constraints between holistic prediction probability vectors and local prediction probability vectors of spatial and spectral components to establish semantic consistency between local and holistic features across spatial and spectral dimensions, thereby enabling the model to learn more discriminative features. Taking the source domain spatial dimension local blocks as an example, given a query image set x^q, we derive classification predictions using both holistic and local

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TABLE IX
CLASS-SPECIFIC CLASSIFICATION ACCURACY (%), OA (%), AA (%), KC (%) ON IP DATASET (FIVE LABELED SAMPLES PER CLASS)

Class	SVM	3DCNN	SSRN	DCFSL	HFSL	Gia-CFSL	FSCF-SSL	RPCL-FSL	MCMD-CFSC
1	69.27	76.24	96.71	95.12	98.54	94.15	**100.00**	99.51	99.51
2	32.56	33.08	41.81	45.83	50.18	43.27	52.42	55.87	**69.66**
3	38.19	35.65	51.28	47.88	63.71	49.64	61.21	62.02	**72.32**
4	52.33	43.07	74.34	78.97	89.96	71.90	**100.00**	86.21	93.23
5	75.73	66.05	83.95	71.84	77.80	72.15	61.92	84.50	**85.75**
6	71.23	72.81	**96.47**	88.17	82.08	79.52	84.83	89.24	94.61
7	89.13	89.40	97.83	98.70	97.83	96.96	**100.00**	97.83	**100.00**
8	62.54	67.81	79.31	81.42	99.39	89.53	**100.00**	90.15	99.32
9	86.67	98.67	**100.00**	99.33	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**
10	44.15	55.49	60.86	59.58	72.23	66.85	**83.04**	74.08	82.11
11	38.98	54.65	45.12	60.17	70.90	66.52	75.06	69.37	**81.50**
12	27.45	34.18	54.66	46.33	68.55	45.80	58.33	68.79	**83.08**
13	85.15	91.52	98.79	94.10	98.95	96.45	**99.50**	97.40	98.20
14	62.82	65.71	80.59	85.53	85.25	83.91	**98.57**	85.37	87.16
15	35.70	49.53	73.33	67.30	90.10	68.77	**100.00**	85.22	93.81
16	90.45	85.21	**98.71**	92.05	98.18	98.18	97.73	97.05	97.73
OA	47.76 ±3.43	53.55 ±3.92	62.31 ±4.42	64.97 ±3.09	73.74 ±3.40	66.45 ±1.60	77.09 ±1.23	74.38 ±1.96	**83.25 ±1.54**
AA	60.15 ±2.38	63.69 ±2.28	77.11 ±2.20	76.20 ±2.44	83.59 ±1.88	76.47 ±0.53	85.79 ±1.45	83.91 ±1.63	**89.87 ±1.09**
KC	41.83 ±3.45	47.69 ±4.15	58.06 ±4.42	60.56 ±3.39	70.37 ±3.67	62.09 ±1.60	73.86 ±2.31	71.11 ±2.14	**81.04 ±1.72**

TABLE X
CLASS-SPECIFIC CLASSIFICATION ACCURACY (%), OA (%), AA (%), KC (%) ON HT DATASET (FIVE LABELED SAMPLES PER CLASS)

Class	SVM	3DCNN	SSRN	DCFSL	HFSL	Gia-CFSL	FSCF-SSL	RPCL-FSL	MCMD-CFSC
1	80.53	**92.81**	80.66	81.82	63.67	81.22	74.07	81.69	81.39
2	75.77	71.74	**83.42**	80.24	55.94	79.37	33.18	78.06	81.78
3	98.06	95.77	98.53	96.83	83.84	93.01	99.60	92.95	**99.52**
4	85.42	87.11	91.56	89.38	81.92	87.95	86.08	86.83	**89.92**
5	89.85	94.61	93.70	96.77	70.72	97.32	**97.63**	94.64	97.51
6	81.19	67.59	80.70	72.66	**83.29**	70.28	82.52	81.19	79.44
7	62.05	49.00	74.79	64.1	80.95	68.53	71.18	**82.59**	79.55
8	25.38	41.48	28.44	39.5	23.29	36.70	23.46	36.41	**41.80**
9	58.95	56.28	51.23	60.86	56.02	62.04	**96.98**	72.39	75.54
10	44.25	58.16	51.92	44.82	54.49	49.18	**60.91**	47.36	52.59
11	50.17	56.50	72.79	52.64	68.88	45.48	55.50	59.61	**76.42**
12	41.48	47.36	58.51	76.51	66.25	70.87	**74.93**	68.41	72.05
13	22.84	28.60	75.47	79.65	67.30	83.05	70.88	**90.84**	77.89
14	96.11	82.96	99.31	83.48	96.23	85.51	96.36	86.19	**99.60**
15	98.03	96.51	**99.58**	94.46	56.03	89.34	80.55	94.23	99.41
OA	63.43 ±2.19	67.49 ±2.88	72.11 ±2.87	71.47 ±1.94	63.96 ±4.19	70.21 ±2.92	70.08 ±3.24	73.60 ±1.39	**77.51 ±1.40**
AA	67.34 ±1.72	68.43 ±2.53	76.04 ±2.39	74.25 ±2.14	67.25 ±4.73	73.08 ±2.50	73.59 ±3.21	76.89 ±1.26	**80.29 ±1.07**
KC	61.59 ±2.32	64.71 ±3.10	69.78 ±3.09	69.20 ±2.08	60.99 ±3.14	67.84 ±3.14	67.64 ±3.59	71.49 ±1.49	**75.69 ±1.49**

branches. The source domain spatial slice self-distillation loss function is formulated as

$$L_{\mathrm{spa}_s}^{\mathrm{self}} = \frac{1}{R}\sum_{i=1}^{I}\sum_{r=1}^{R} H\left(\boldsymbol{p}_{q_{i,r_s}}^{\mathrm{spa}}\boldsymbol{p}_{q_{is}}\right) \tag{11}$$

where $H(\cdot,\cdot)$ denotes the cross-entropy loss, R represents the number of spatial and spectral blocks, and I represents the total number of query set samples in each episode. However, enforcing consistency solely between local and holistic prediction probabilities within the same sample may lead to a critical limitation: the model might adopt an oversimplified learning strategy, focusing merely on achieving local–holistic prediction alignment within individual samples while failing to extract essential category-level features shared across different samples. This could potentially result in suboptimal model performance and poor generalization to unseen data. To address this challenge, we introduce cross-sample distillation between blocks of x_{q_i} and another randomly selected query image x_{q_j} from the same category. This approach establishes semantic consistency among intraclass samples, compelling the model to learn consistent representations across different images within the same category and capture more comprehensive and

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TABLE XI
CLASS-SPECIFIC CLASSIFICATION ACCURACY (%), OA (%), AA (%), KC (%) ON SA DATASET (FIVE LABELED SAMPLES PER CLASS)

Class	SVM	3DCNN	SSRN	DCFSL	HFSL	Gia-CFSL	FSCF-SSL	RPCL-FSL	MCMD-CFSC
1	96.71	96.13	98.11	98.96	99.35	99.27	**100.00**	99.44	99.97
2	92.20	97.06	89.49	98.48	94.22	99.60	97.58	99.12	**99.65**
3	87.85	93.09	90.64	96.97	96.54	94.40	**99.54**	97.72	98.62
4	99.19	98.00	99.07	99.73	99.84	98.90	**100.00**	99.63	99.88
5	95.66	92.89	95.07	91.93	93.33	87.75	**97.27**	91.80	96.58
6	98.11	98.77	99.20	99.32	99.33	98.78	99.82	98.42	**99.83**
7	97.53	98.97	99.89	99.11	97.67	99.26	**100.00**	99.47	99.87
8	56.29	68.47	69.57	82.73	79.11	83.56	77.17	85.49	**89.69**
9	93.75	97.50	99.50	99.28	99.27	97.85	**100.00**	99.76	99.99
10	74.74	77.81	86.41	86.94	**93.63**	83.18	93.43	89.53	91.98
11	92.08	90.19	96.71	95.61	99.72	91.46	**99.91**	97.10	99.51
12	98.48	99.43	99.32	99.14	95.11	97.79	96.10	99.46	**99.79**
13	95.81	94.26	98.60	98.56	98.08	95.12	**100.00**	97.07	98.38
14	91.14	93.44	96.83	98.72	96.87	97.70	96.53	98.21	**99.37**
15	62.16	63.57	69.17	72.16	71.58	70.57	**91.33**	78.33	88.28
16	77.16	82.99	92.91	90.89	97.52	86.59	90.57	93.31	**99.97**
OA	80.70 ±3.16	84.74 ±1.41	86.76 ±3.15	90.55 ±1.04	89.92 ±2.10	89.50 ±0.92	92.82 ±1.86	92.29 ±0.90	**95.32 ±0.68**
AA	88.05 ±1.45	90.16 ±1.36	92.53 ±1.94	94.28 ±0.97	94.45 ±1.45	92.61 ±0.99	96.20 ±0.83	95.24 ±0.92	**97.31 ±0.61**
KC	78.62 ±3.41	83.03 ±1.57	85.29 ±3.43	89.48 ±1.14	88.79 ±2.32	88.31 ±1.02	92.03 ±2.03	91.47 ±1.00	**94.78 ±0.76**

TABLE XII
CLASS-SPECIFIC CLASSIFICATION ACCURACY (%), OA (%), AA (%), KC (%) ON LK DATASET (FIVE LABELED SAMPLES PER CLASS)

Class	SVM	3DCNN	SSRN	DCFSL	HFSL	Gia-CFSL	FSCF-SSL	RPCL-FSL	MCMD-CFSC
1	84.92	88.14	94.95	99.33	97.10	98.80	98.30	98.48	**99.51**
2	57.88	38.01	72.19	94.75	97.42	91.48	97.32	93.14	**97.78**
3	68.44	77.06	92.68	92.78	90.04	91.14	88.18	93.43	**94.84**
4	56.97	67.98	77.52	90.48	85.95	88.75	87.67	91.75	**95.97**
5	69.90	68.26	94.06	93.58	96.79	92.91	**98.09**	88.30	97.44
6	82.75	86.26	95.87	92.94	94.54	92.84	95.42	93.78	**97.33**
7	**99.92**	99.87	99.78	99.84	99.27	99.79	99.68	98.96	99.09
8	77.30	72.03	65.19	81.26	88.68	83.43	**95.34**	85.82	86.50
9	50.48	45.86	75.00	82.10	**91.50**	79.22	90.75	88.37	86.71
OA	78.27 ±5.53	81.39 ±2.52	88.67 ±3.73	94.92 ±1.53	93.68 ±2.44	94.11 ±1.92	94.81 ±1.96	95.09 ±1.28	**97.19 ±0.92**
AA	72.06 ±3.47	71.50 ±2.02	85.25 ±4.51	91.89 ±2.20	93.48 ±2.14	90.93 ±1.45	94.53 ±1.03	92.45 ±1.32	**95.02 ±1.18**
KC	72.81 ±6.48	76.35 ±2.98	85.52 ±4.60	93.38 ±1.94	91.84 ±3.08	92.35 ±2.44	93.28 ±2.49	93.60 ±1.63	**96.32 ±1.18**

nuanced feature representations. The source domain spatial block cross-sample distillation loss is formulated as

$$L_{\mathrm{spa}_s}^{\mathrm{cross}} = \frac{1}{R}\sum_{i=1}^{I}\sum_{r=1}^{R} H\left(\boldsymbol{p}_{q_{i,r_s}}^{\mathrm{spa}}, \boldsymbol{p}_{q_{j_s}}\right), \quad j \neq i. \tag{12}$$

This strategy drives the model to discover generalizable features that are consistent across different samples within the same category, rather than relying on sample-specific patterns.

In the metalearning paradigm, training data in each episode constitutes scenario tasks comprising randomly sampled images from different categories, resulting in distinct combinations across episodes. This episodic variation leads to substantial semantic space fluctuations, requiring the network to continuously adapt to different visual pattern combinations, which impedes the accumulation of cross-episode knowledge. To overcome this limitation, we propose a cross-episode knowledge distillation strategy. During meta-training, we implement the update of the local branch feature extractor f_l as the EMA of the global branch feature extractor f_φ. The local branch parameter update is formulated as

$$\theta_l \leftarrow m\theta_l + (1-m)\,\theta_\varphi \tag{13}$$

where θ_l and θ_φ denote the model parameters of the f_l and f_φ, and $m \in [0, 1]$ represents the momentum controlling the update rate. The spatial dimension block loss of the source domain is defined as

$$L_{\mathrm{spa}}^{s} = L_{\mathrm{spa}_s}^{\mathrm{self}} + L_{\mathrm{spa}_s}^{\mathrm{cross}}. \tag{14}$$

The complete multidimensional distillation loss is formulated as

$$L_{\mathrm{md}} = L_{\mathrm{spa}}^{s} + L_{\mathrm{spe}}^{s} + L_{\mathrm{spa}}^{t} + L_{\mathrm{spe}}^{t}. \tag{15}$$

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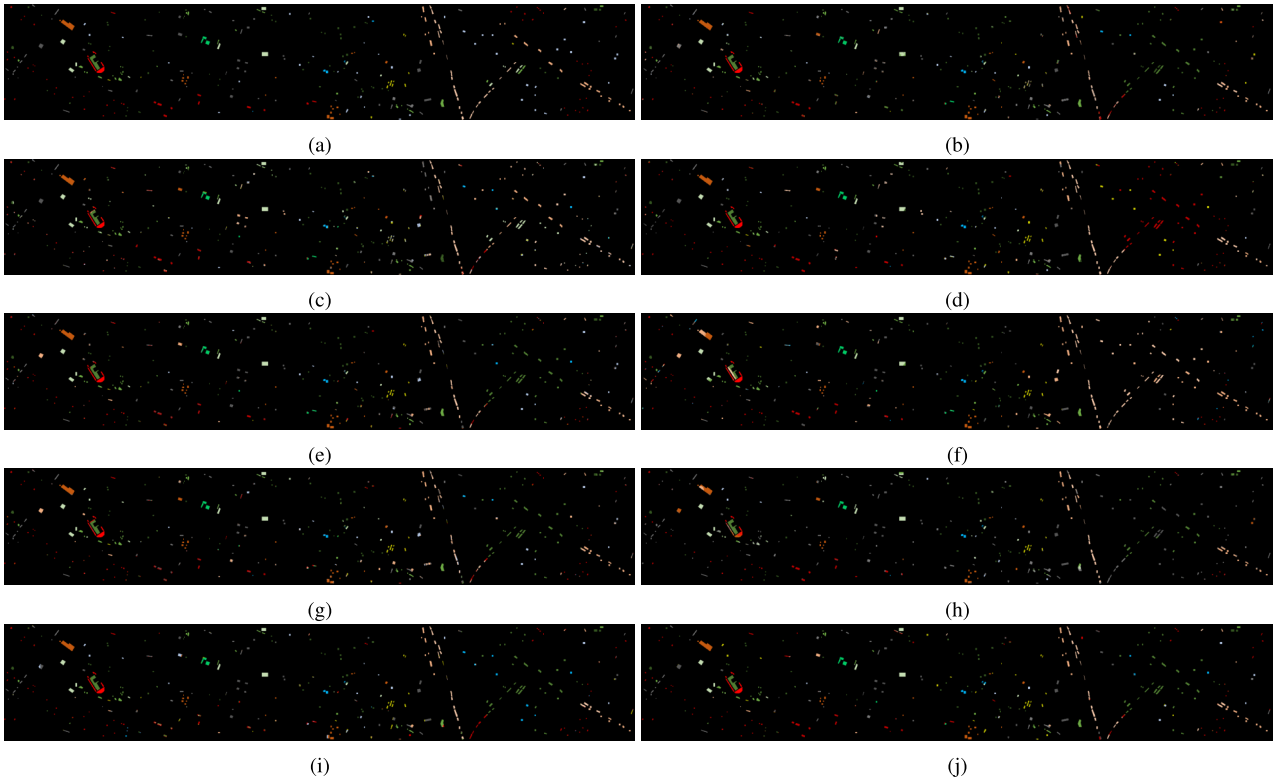


Fig. 11. Classification maps on the HT dataset. (a) Ground-truth. (b) SVM. (c) 3DCNN. (d) SSRN. (e) DCFSL. (f) HFSL. (g) Gia-CFSL. (h) FSCF-SSL. (i) RPCL-FSL. (j) MCMD-CFSC. Enlarge the images to see the difference.

The overall loss for the MCMD-CFSC is defined as the sum of three losses

$$L_{\text{total}} = L_{\text{fsl}} + \lambda_1 L_{\text{mc}} + \lambda_2 L_{\text{md}} \tag{16}$$

where λ_1 and λ_2 are regularization parameters.

IV. Experiments

A. Experimental Datasets

To comprehensively assess the proposed method's effectiveness, we selected five prominent HSI datasets: Chikusei (CH), Indian Pines (IP), Houston (HT), Salinas (SA), and LongKou (LK) [61]. Following the FSL experimental framework from [32], we used CH as the source domain and the remaining four datasets as target domains.

1) Source Domain Dataset:

A. Chikusei (CH): The CH dataset was acquired using the Headwall Hyperspec-VNIR-C sensor, featuring dimensions of 2517 × 2335 pixels and a spatial resolution of 2.5 m/pixel. It includes 19 diverse land cover categories and comprises 128 spectral bands with a wavelength range of 343–1018 nm. Table I presents the sample count for each category, while Fig. 6 displays the pseudo-color image of the CH dataset alongside its corresponding ground-truth map.

2) Target Domain Datasets:

A. Indian pines: The IP dataset was collected using the AVIRIS sensor at the test site in northwestern Indiana. It consists of 145 × 145 pixels with a low spatial resolution of 20 m/pixel. This dataset features 16 land cover categories and provides extensive spectral information across 200 bands, ranging from 400 to 2500 nm. The number of samples in each category is detailed in Table II. In addition, Fig. 7 displays the pseudo-color image of the IP dataset alongside the corresponding ground-truth map.

B. Houston: The HT dataset was collected in 2012 by an airborne sensor around HT University. It is a large dataset with 349 × 1905 pixels and a spatial resolution of 2.5 m/pixel. It encompasses 15 land cover categories and features 144 spectral bands with wavelengths ranging from 364 to 1046 nm. Table III presents the sample counts for the training and test sets in each category. In addition, Fig. 8 displays the pseudo-color image of the HT dataset along with the ground-truth maps for both the training and test sets.

C. Salinas: The SA dataset captured by the AVIRIS sensor at the SA Valley test site in California consists of 512 × 217 pixels with a spatial resolution of 3.7 m/pixel. It includes 16 land cover categories and features 204 spectral bands spanning wavelengths from 400 to 2500 nm. Table IV lists the number of samples for each category, while Fig. 9 displays the pseudo-color image of the SA dataset alongside the corresponding ground-truth map.

D. WHU-Hi-LongKou: The LK dataset was obtained from the Headwall Nano-Hyperspec sensor mounted on an autonomous aerial vehicles (AAV) in Longkou Town, Jingzhou City, Hubei, China. It features a 550 × 400 pixel image with a spatial resolution of 0.463 m/pixel. The dataset includes nine land cover categories and 270 spectral bands spanning wavelengths from 400 to 1000 nm. Table V outlines the sample counts for each category, while Fig. 10 presents a pseudo-color image of the LK dataset alongside the corresponding ground-truth map.

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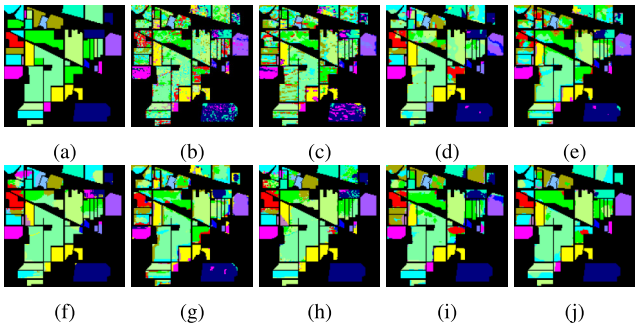


Fig. 12. Classification maps on IP dataset. (a) Ground-truth. (b) SVM. (c) 3DCNN. (d) SSRN. (e) DCFSL. (f) HFSL. (g) Gia-CFSL. (h) FSCF-SSL. (i) RPCL-FSL. (j) MCMD-CFSC. Enlarge the images to see the difference.

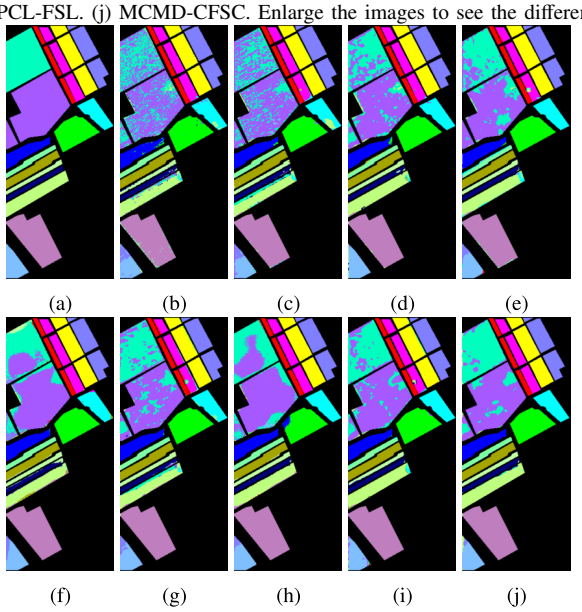


Fig. 13. Classification maps on the SA dataset. (a) ground-truth. (b) SVM. (c) 3DCNN. (d) SSRN. (e) DCFSL. (f) HFSL. (g) Gia-CFSL. (h) FSCF-SSL. (i) RPCL-FSL. (j) MCMD-CFSC. Enlarge the images to see the difference.

B. Experimental Setups

The workstation used in the experiment has the following hardware configuration: Intel Xeon W7-3445 CPU (2.9 GHz), 256-GB memory, and NVIDIA RTX6000 Ada GPU (48 GB). The experimental code is implemented using the open-source framework PyTorch.

In the MCMD-CFSC method, the patch block size is set to $9 \times 9 \times B$, with B representing the number of bands. The training process utilizes both source and target domain data simultaneously across 5000 iterations. Here, K is set to 1, indicating that one sample per class serves as the support set, while each category in the query set employs 19 samples for training. The temperature coefficient τ is 0.05. The model is optimized using the stochastic gradient descent (SGD) optimizer with a learning rate of 0.001 and a weight decay of 0.0001. Classification performance is assessed through overall accuracy (OA), average accuracy (AA), and the kappa coefficient (KC).

C. Parameter Tuning

To optimize model performance on the IP, HT, SA, and LK datasets, we performed parameter tuning for the momentum m in the MD module, the L_{mc} loss weight λ_1, and the L_{md} loss weight λ_2. The momentum m was selected from {0.9, 0.93, 0.95, 0.97, 0.99}, the L_{mc} loss weight λ_1 was selected from {0.1, 0.5, 1, 1.5, 2}, and the L_{md} loss weight λ_2 was selected from {0.1, 0.5, 1, 1.5, 2}. To alleviate the pressure of parameter optimization across different datasets, we adopted a suboptimal parameter adjustment strategy. This approach involved sequential parameter tuning, where each parameter was optimized independently while maintaining others at their default values. Once a parameter was optimized, it was fixed before proceeding to the next parameter. To ensure statistical reliability, we conducted ten independent trials for each parameter combination and used their averaged OA as the evaluation metric. Tables VI–VIII present the performance variations across the IP, HT, SA, and LK datasets under different values of momentum m, L_{mc} loss weight λ_1, and L_{md} loss weight λ_2. Based on these comprehensive experiments, we established the optimal parameters for each dataset. The momentum coefficient m was determined to be 0.95, 0.90, 0.93, and 0.90 for IP, HT, SA, and LK datasets, respectively. The loss weight λ_1 was set to 2, 0.1, 2, and 0.1 for the respective datasets, while λ_2 was consistently set to 1 across all datasets.

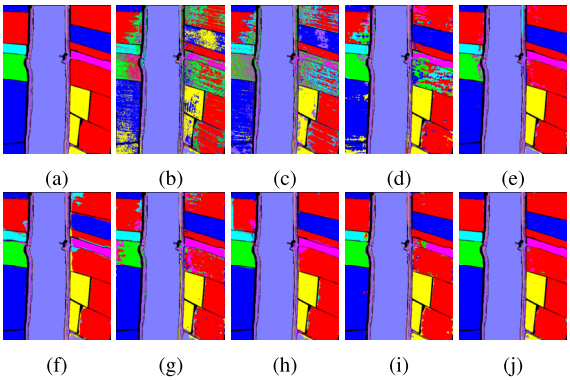


Fig. 14. Classification maps on LK dataset. (a) ground-truth. (b) SVM. (c) 3DCNN. (d) SSRN. (e) DCFSL. (f) HFSL. (g) Gia-CFSL. (h) FSCF-SSL. (i) RPCL-FSL. (j) MCMD-CFSC. Enlarge the images to see the difference.

D. Comparison With Other Methods

In situations with limited labeled samples, we assessed the classification performance of the proposed MCMD-CFSC method by comparing it with several representative methods. These methods include traditional machine learning methods such as SVM [8], traditional deep learning methods such as 3DCNN [16] and SSRN [17], as well as cross-domain FSL methods including DCFSL [32], HFSL [62], Gia-CFSL [35], FSCF-SSL [63], and RPCL-FSL [64]. The training and testing procedures for these methods are as follows. To ensure a fair comparison, we maintain consistency in classes between the training and testing for traditional deep learning methods, 3DCNN, and SSRN. We randomly select five labeled samples from each class in the target domain dataset to create the training set, with the rest forming the test set. Similarly, for the cross-domain FSL methods DCFSL, HFSL, Gia-CFSL, FSCF-SSL, RPCL-FSL, and the proposed MCMD-CFSC, we extract five labeled samples from each class in the target domain

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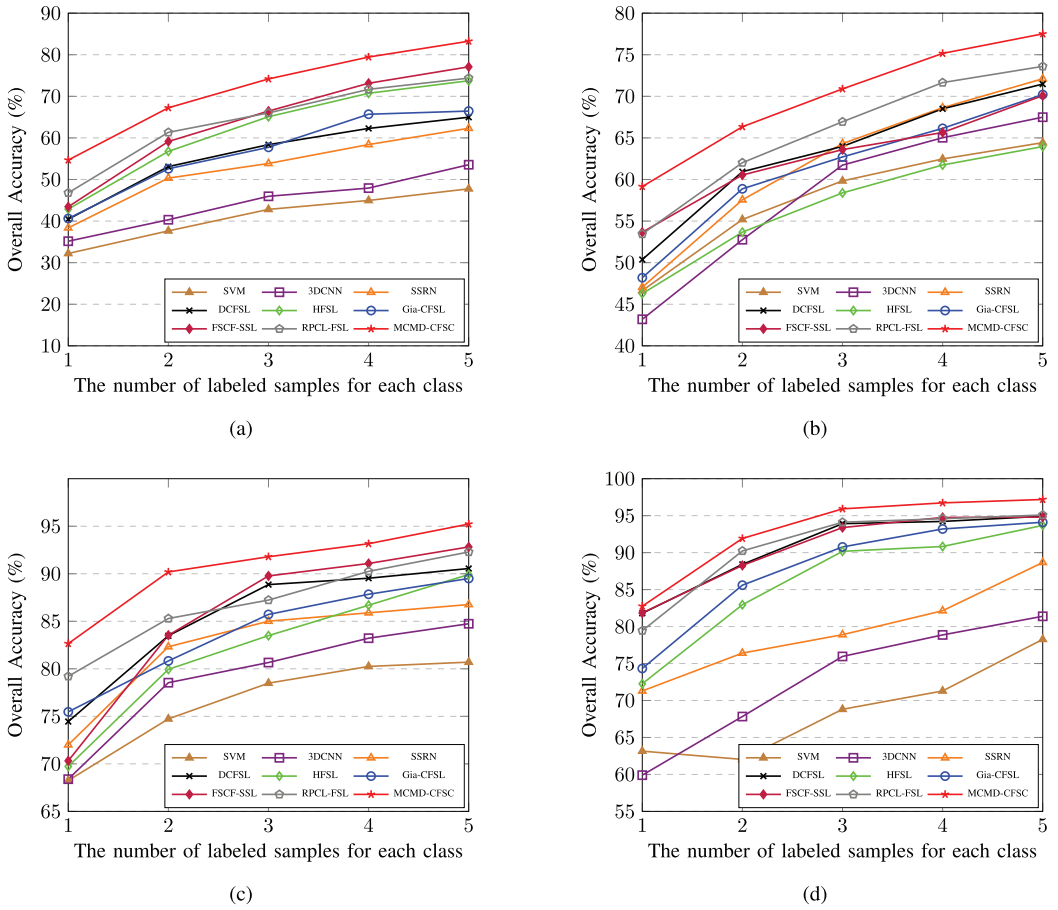


Fig. 15. OA (%) of all methods under different numbers of labeled samples per class in the target domain dataset: (a) IP, (b) HT, (c) SA, and (d) LK.

TABLE XIII
ABLATION COMPARISON RESULTS (%) OF EACH VARIANT OF MCMD-CFSC

Module				Dataset			
FS	MC	MD-SPA	MD-SPE	IP	HT	SA	LK
✓				76.21	74.86	92.39	96.31
✓	✓			82.76	76.31	94.82	96.55
✓	✓	✓		82.99	76.76	95.12	97.02
✓	✓	✓	✓	**83.25**	**77.51**	**95.32**	**97.19**

for training, reserving the remaining samples for testing. For HFSL and FSCF-SSL methods, the mini-ImageNet training set serves as the source domain for training the base classes.

For the DCFSL, Gia-CFSL, RPCL-FSL, and MCMD-CFSC methods, we use the CH dataset as the source domain, randomly selecting 200 samples from each base class for training. For the HT dataset, which has predetermined training and testing sets, we randomly choose five labeled samples from each class in the training set and test on all samples in the test set. To reduce randomness in sampling, all experiments are conducted ten times, and the average results are reported. This approach minimizes the influence of random factors on accuracy and reliability, ensuring the evaluation truly reflects each method's performance in few-sample classification scenarios. This rigorous experimental design allows for a fair and thorough assessment of the MCMD-CFSC method's performance.

Tables IX–XII present comparative results across methods using the IP, HT, SA, and LK datasets, each containing five labeled samples per class. As indicated in Table IX, the proposed MCMD-CFSC method significantly surpasses all other methods on the IP dataset. Specifically, it outperforms the cross-domain FSL methods FSCF-SSL and RPCL-FSL in OA, AA, and KC, showing improvements of 6.16%, 4.08%, and 7.18% over FSCF-SSL, and 8.87%, 5.96%, and 9.93% over RPCL-FSL. Furthermore, MCMD-CFSC excelled in classifying fine categories like soybeans and corn, with accuracy increases exceeding 10% for classes 2, 3, 11, and 12 compared to other top methods. This notable performance enhancement underscores the effectiveness and superiority of the proposed method, demonstrating its efficiency in HSI classification and its potential in cross-domain few-shot classification challenges. Table X presents the comparative classification results on the HT dataset. Among all evaluated methods, the traditional deep learning approach, SSRN, and the cross-domain FSL method RPCL-FSL demonstrate notable performance. However, our proposed MCMD-CFSC method consistently surpasses these methods across all evaluation metrics. Specifically, in terms of OA, MCMD-CFSC achieves 77.51%, surpassing SSRN by 5.4% and RPCL-FSL by 3.91%, demonstrating its superior capability in comprehensive classification tasks. Moreover, MCMD-CFSC exhibits exceptional performance in several

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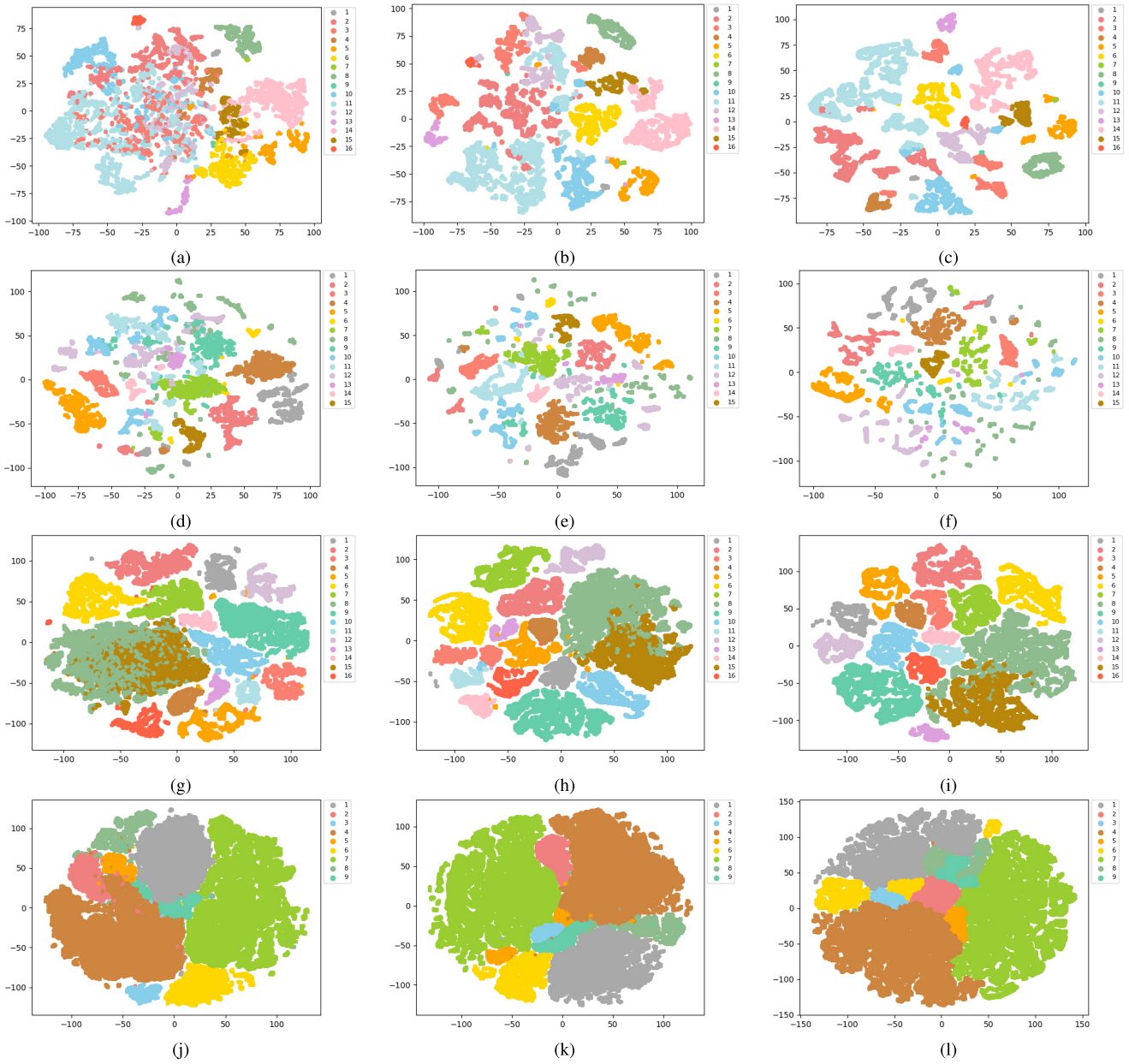


Fig. 16. 2-D feature visualization on four datasets. (a) Gia-CFSL features on IP. (b) RPCL-FSL features on IP. (c) MCMD-CFSC features on IP. (d) Gia-CFSL features on HT. (e) RPCL-FSL features on HT. (f) MCMD-CFSC features on HT. (g) Gia-CFSL features on SA. (h) RPCL-FSL features on SA. (i) MCMD-CFSC features on SA. (j) Gia-CFSL features on LK. (k) RPCL-FSL features on LK. (l) MCMD-CFSC features on LK.

specific categories, particularly in categories 3, 5, 14, and 15, where the classification accuracy approaches or reaches 99%. These results comprehensively validate the effectiveness and superiority of our method in addressing HSI classification challenges. Table XI presents the classification performance comparison on the SA dataset, where the proposed MCMD-CFSC method demonstrates superior performance across all metrics. The method achieves notable improvements over the state-of-the-art cross-domain FSL method FSCF-SSL, with gains of 2.5%, 1.11%, and 2.75% in OA, AA, and KC, respectively. When compared to PRCL-FSL, MCMD-CFSC shows even more substantial improvements of 8.56%, 4.78%, and 9.49% in these metrics. The method's effectiveness is particularly evident in its outstanding performance on category 8, which is considered the most challenging classification task in the dataset, further validating the robustness of our approach.

Table XII presents the classification results of various methods on the LK dataset, where the proposed MCMD-CFSC method consistently outperforms all comparison methods. Specifically, in terms of the three evaluation metrics of OA, AA, and KC, the MCMD-CFSC method surpasses the best-performing cross-domain FSL method FSCF-SSL by 2.41%, 1.29%, and 2.66%, respectively, and demonstrates significant improvements over the best-performing traditional deep learning method SSRN by 8.47%, 4.96%, and 9.4%, respectively.

To illustrate the classification performance when only five labeled samples per category are available in the target domain, we present the classification maps generated by all compared methods on the IP, HT, SA, and LK datasets, alongside their

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TABLE XIV
COMPUTATION TIME (S), FLOPS (M), AND PARAMETERS (M) ON EACH DATASET WITH ALL METHODS

Dataset	Metric	Method								
		SVM	3DCNN	SSRN	DCFSL	HFSL	Gia-CFSL	FSCF-SSL	RPCL-FSL	MCMD-CFSC
IP	Training time	0.86	45.20	66.28	1781.30	639.04	3796.71	1195.40	473.85	3662.06
	Testing time	0.09	1.768	8.41	1.06	7.23	1.24	6.65	1.21	3.07
	FLOPs	-	0.20	1.38	47.28	400.45	130.82	400.45	47.37	29.07
	#Params	-	0.10	0.35	4.28	6.27	0.86	6.28	4.27	0.31
HT	Training time	0.85	41.23	78.04	1617.51	546.15	3102.85	967.72	376.84	3277.50
	Testing time	0.07	1.92	9.99	1.57	5.72	1.18	4.16	1.53	3.60
	FLOPs	-	0.14	1.04	47.28	399.94	124.86	399.94	46.92	28.8
	#Params	-	0.07	0.26	4.28	6.19	0.85	6.19	4.26	0.30
SA	Training time	0.84	51.17	127.77	1353.44	846.61	3902.90	2699.09	567.68	3860.39
	Testing time	0.37	10.88	46.92	5.70	28.57	5.37	27.02	5.56	27.95
	FLOPs	-	0.21	1.41	47.31	400.48	130.85	400.48	47.40	29.09
	#Params	-	0.10	0.35	4.28	6.28	0.86	6.28	4.27	0.31
LK	Training time	0.99	50.18	337.75	1310.75	1306.04	2782.54	2658.61	661.10	3452.89
	Testing time	1.08	44.99	183.96	25.15	1306.04	23.68	144.14	25.53	61.24
	FLOPs	-	0.27	1.81	47.85	401.08	93.12	401.08	47.94	29.42
	#Params	-	0.14	0.45	4.28	6.38	0.86	6.38	4.28	0.32

corresponding ground-truth maps. As depicted in Figs. 11–14, our proposed MCMD-CFSC method produces classification maps that exhibit the highest visual consistency with the ground-truth, showing notably reduced interclass confusion compared to other methods.

To comprehensively evaluate the MCMD-CFSC method's performance across varying numbers of labeled samples, we conducted a series of systematic experiments. In the training phase, we randomly selected 1–5 labeled samples per category from the target domain, with the remaining samples designated as the test set to simulate real-world scenarios with limited labeled data. The reported classification results represent the average of ten independent runs, ensuring statistical reliability. This experimental protocol demonstrates both the robustness of the MCMD-CFSC method in few-shot scenarios and its effectiveness in HSI classification tasks. Fig. 15 illustrates the OAs of all methods across the IP, HT, SA, and LK datasets using 1–5 shot samples per category. The MCMD-CFSC method consistently achieves superior performance across all datasets and sample sizes, demonstrating its robust capability to adapt to diverse data distributions and scene characteristics while effectively capturing class-specific features across different datasets. To visualize the feature extraction capabilities of different methods, we employed t-distributed stochastic neighbor embedding (t-SNE) dimensionality reduction to project features into a 2-D space. Fig. 16 presents the t-SNE visualizations of three cross-domain FSL methods (Gia-CFSL, RPCL-FSL, and MCMD-CFSC) on the IP, HT, SA, and LK datasets. Compared to Gia-CFSL and RPCL-FSL, our MCMD-CFSC method exhibits more distinct class boundaries and larger interclass separations across all datasets, indicating enhanced feature discriminability.

In the IP dataset, particularly challenging classes include the spectrally similar "soybean notill" (tenth class), "soybean mintill" (11th class), and "soybean clean" (12th class). As shown in Fig. 16(a)–(c), MCMD-CFSC establishes clear decision boundaries among these classes, significantly reducing their confusion. Similarly, for the SA dataset, where the "grapes untrained" (eighth class) and "vineyard untrained" (15th class) share similar spectral characteristics, Fig. 16(g)–(i) demonstrates MCMD-CFSC's ability to effectively distinguish between them. Furthermore, Fig. 16(d)–(f) and (j)–(l) shows that MCMD-CFSC achieves the lowest class confusion on the HT and LK datasets. These results demonstrate that MCMD-CFSC effectively reduces intraclass feature variations while enhancing interclass separability, leading to more robust and accurate classification performance in complex HSI classification tasks.

E. Ablation Study

The main contributions of MCMD-CFSC lie in the MC and MD methods. To better analyze the effectiveness of these two methods, we conducted ablation studies to evaluate their contributions. As shown in Table XIII, we focused on four scenarios: 1) FSL only in source and target domains; 2) adding MC to scenario 1); 3) adding MD-SPA to scenario 2); and 4) adding MD-SPE to scenario 3), where MD-SPA represents holistic–local consistency constraints in the spatial dimension, and MD-SPE represents holistic–local consistency constraints in the spectral dimension.

Table XIII reveals that incorporating MC into the baseline scenario improved OA by 6.55%, 1.45%, 2.43%, and 0.24% on the IP, HT, SA, and LK datasets, demonstrating MC's ability to leverage textual prior knowledge for image prototype correction. Adding MD-SPA further enhanced performance by 0.23%, 0.45%, 0.3%, and 0.47%, respectively, indicating effective learning of spatial holistic–local relationships. The subsequent integration of MD-SPE yielded additional improvements of 0.26%, 0.75%, 0.2%, and 0.17%, confirming the model's capability to capture spectral holistic–local relationships.

F. Computational Complexity Analysis

As shown in Table XIV, we evaluated the computational complexity of models through training time, testing time,

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floating-point operations (FLOPs), and parameter count. For traditional machine learning (SVM) and conventional deep learning methods (3DCNN and SSRN), the training time reflects single-domain training with limited labeled samples. In contrast, cross-domain FSL methods, including DCFSL, Gia-CFSL, RPCL-FSL, and our proposed MCMD-CFSC, require both source domain training and target domain fine-tuning time. In addition, HFSL and FSCF-SSL methods necessitate extra model transfer time.

Analysis of Table XIV indicates that while cross-domain FSL methods achieve superior classification performance compared to conventional deep learning approaches, they demand higher computational resources in terms of training time and FLOPs. Notably, our proposed MCMD-CFSC outperforms Gia-CFSL while maintaining comparable time costs. MCMD-CFSC achieves superior classification accuracy with reduced FLOPs and parameters, enhancing its practicality for real-world applications. These characteristics establish MCMD-CFSC as an efficient solution for cross-domain few-shot HSI classification tasks.

V. Conclusion

The proposed MCMD-CFSC method implements a multimodal prototype correction strategy by utilizing pre-trained language models to extract spatial and spectral text features from corresponding text descriptions, and achieves precise alignment between image spatial prototypes and spatial text features, as well as between spectral prototypes and spectral text features. This multimodal prototype correction approach effectively mitigates interclass similarity interference and significantly enhances the model's capability to distinguish various ground objects and land cover types. Furthermore, the dual knowledge distillation framework strengthens the consistency between local and global semantics, making feature representations more discriminative. Experimental results confirm the effectiveness of this method across different scenarios and datasets, demonstrating its superior performance and practical application potential. However, the method still has some limitations. The method is highly dependent on high-quality text descriptions. In specialized domains lacking expert prior knowledge, obtaining accurate categorical semantic descriptions is challenging, which may affect the method's widespread application.

Therefore, future research will be dedicated to exploring automated text description generation techniques to reduce the degree of dependence on expert knowledge. From a practical impact perspective, the spatial and spectral text description method proposed in this study provides researchers with a design template for text descriptions applicable to hyperspectral data, inspiring subsequent researchers to develop text description schemes that better align with hyperspectral data characteristics. The prototype correction mechanism offers new insights for using multimodal data to optimize image prototypes, while the dual knowledge distillation strategy provides a framework for exploring more discriminative feature representations.

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